

Conquer

Asset Specifications & Pixelart Style Guide
Version 2.0 — Pixelart Edition — May 2026

This is the canonical reference for all visual assets in Conquer. The game uses a **pixelart aesthetic** — specifically a 16-bit-era SNES-style with a curated 32-color palette and crisp pixel-perfect sprites. This document defines: dimensions, color palette, animation conventions, AI generation prompts, file organization, and the re-skinnable sprite-set architecture that allows future style alternatives.

Production order: terrain (1 day) → 1 sample monster + 1 building (1 day) → verify style works in-engine → expand category by category. Don't try to generate all 330 sprites in one batch — iterate per category and refine the style guide as you learn what works.

1. Art Direction

The Conquer aesthetic

Conquer is a **pixelart strategy MMO** in the spirit of 16-bit JRPGs and modern indie pixel-strategy games (Kingdom Two Crowns, Songs of Conquest, Mindustry). The style should feel:

Aspect	Direction
Era reference	SNES / 16-bit (1990-1995), not NES (8-bit) and not modern HD-pixel
Detail level	Readable at native resolution, charming when zoomed, never busy
Color palette	32 colors max — limited palette enforces visual cohesion
Outline style	Soft 1px outlines (NOT hard black borders like comic style)
Anti-aliasing	NONE on sprites — crisp pixel edges only
Lighting	Top-left light source, soft shadows on opposite side
Mood	Warm fantasy — not grimdark, not saccharine. Adventurous.
Reference games	Kingdom Two Crowns, Songs of Conquest, Mindustry, Stardew Valley UI

What we're NOT doing

Style	Reason for exclusion
3D rendered (Lords Mobile, ROK)	Not what we are — different aesthetic, different production cost
Hand-drawn 2D (Tribal Wars)	Skipped for v1 — too production-heavy for solo dev
8-bit NES style	Too retro, too pixely, harder to convey fantasy detail
HD pixel art (Octopath, Owlboy)	Too production-heavy — beautiful but 5-10× more work
Vector / SVG	Doesn't fit pixelart commitment
Photo-realistic	Doesn't fit indie/strategic-MMO niche we're targeting

Why pixelart for this project

Production-friendly: Solo developer can produce + iterate sprites quickly. AI tools (Stable Diffusion, Midjourney) generate decent pixel-art drafts that need minimal cleanup. Aseprite (~\$20 once) is the only tool needed.

Performance-friendly: Tiny PNG file sizes (1-4 KB per sprite). Bandwidth-cheap for shared-hosting MMO with 1000s of sprite requests per session.

Style-cohesive: Limited 32-color palette enforces visual consistency. New sprites added late in development still match the early ones.

Genre-distinctive: Most major strategy MMOs are 3D rendered. Pixelart sets Conquer apart visually and signals 'made with care' rather than 'mass-produced'.

2. The Conquer 32-Color Palette

Every sprite MUST use only colors from this palette. Inspired by the AAP-64 and DB32 community palettes, customized for fantasy strategy. The palette is grouped by purpose (skin, stone, wood, grass, water, fire, magic, gold) so artists know which range to draw from for any element.

 #f4d6a0 Skin Light	 #c89571 Skin Mid	 #8c5a3c Skin Dark	 #2a1a14 Hair Black	 #d6d4cc Stone Light	 #9c9a8e Stone Mid	 #5e5b50 Stone Dark	 #2a2823 Stone Black
 #c89968 Wood Light	 #8c6437 Wood Mid	 #5e3f1f Wood Dark	 #3a2410 Wood Deep	 #a4d65e Grass Light	 #6ba73a Grass Mid	 #3d6b1e Grass Dark	 #1f3d10 Grass Deep
 #7cc4e0 Water Light	 #3d8ab0 Water Mid	 #1f5878 Water Dark	 #0e2c40 Water Deep	 #ffb84a Fire Light	 #e0631a Fire Mid	 #9c2e10 Fire Dark	 #c97cff Magic Light
 #7c3da8 Magic Mid	 #3d1f5e Magic Dark	 #ffe66a Gold Light	 #d4a020 Gold Mid	 #7c5e10 Gold Dark	 #fafafa White	 #0a0a14 Black	 #5ed4d6 UI Cyan

Palette download

The palette ships as **data/palette/conquer-32.gpl** (GIMP/Aseprite format). Import in Aseprite: Edit → Preferences → Color → Load Palette → choose conquer-32.gpl. All new sprites should be created with this palette active to enforce the rule automatically.

Palette use guidelines

Element	Palette range to use
Player skin	Skin Light → Skin Mid → Skin Dark + Hair Black for shadows
Player city / building stone	Stone Light → Stone Mid → Stone Dark + Stone Black
Wooden buildings, palisades	Wood Light → Wood Dark, accent with Stone for trim
Terrain — grass	Grass Light → Grass Deep — vary 2-3 colors per tile
Terrain — water	Water Light → Water Deep, animate with Light↔Mid wave cycles
Fire effects, red dragon, lava	Fire Light → Fire Dark + Gold for hottest cores
Magic effects, undead aura, gems	Magic palette + UI Cyan for cool magic
Treasure, gold mine, royalty trim	Gold Light → Gold Dark + Wood for backgrounds
UI elements	Stone palette for frames + UI Cyan for accents + Gold for highlights

The 'forbidden' colors rule

If you find yourself wanting to add a color outside this palette, **stop**. Pick the closest existing color. The constraint is the feature — it's what makes 300 sprites by different artists look like one game. Adding 'just one more color' is how palettes balloon to 500 colors and the game looks like a hodgepodge.

3. Sprite Size Reference

All sprites are pixel-perfect at the dimensions below. **No anti-aliasing.** PNG-32 with transparent background. File names lowercase with underscores.

Element	Native Size	Footprint	Notes
Terrain Tile	32 × 32 px	1×1 tile	Smaller than v1 spec — better mobile readability
Field Object (Resource Mine, etc.)	32 × 32 px	1×1 tile	1 sprite per type × 3 visual tiers
Standard Monster (map)	32 × 32 px	1×1 tile	Orc, Skeleton, Golem, Goblin, Deathkar
Dragon (map)	48 × 48 px	Overlaps adjacent tiles	Visually larger to signal danger
Magdar (map)	64 × 64 px	Overlaps 2×2 area	Largest map sprite
Player City	64 × 64 px	2×2 tiles	4 visual tiers (L1-5, 6-15, 16-25, 26-30)
Shrine (C/B/A)	128 × 128 px	4×4 tiles	Pixelart at this size feels grand
Congress (S)	128 × 128 px	4×4 tiles	Centerpiece of map
Battle Sprite (Monster)	64 × 64 px (Std) 96 × 96 px (Drachen) 128 × 128 px (Magdar)	Battle screen	2× map size — readable in battle UI
Battle Sprite (Troop)	48 × 48 px	Battle screen	5 troops × 3 types
Treasure Icon	48 × 48 px	UI	77 unique designs across 5 grades
Charm Icon	32 × 32 px	UI	24 designs (8 stats × 3 grades)
Building Icon (City View)	96 × 96 px	City screen	13 buildings × 4 visual tiers
Resource Icon	16×16 + 32×32 px	UI	Both sizes for HUD and detail views
Action Button Icon	32 × 32 px	UI	Build, Research, Attack, etc.
VIP Crown Icon	32 × 32 px	UI	20 levels (or 4 tiers)
Speedup Icon	32 × 32 px	UI	Build/Research/Train/Recover/Generic

Display zoom levels

Pixelart only scales cleanly in integer multiples. The renderer must use **nearest-neighbor scaling** (CSS: image-rendering: pixelated) and only zoom by 1×, 2×, 3×, 4×. Fractional zooms produce blurry output — disable them.

Zoom	Tile size on screen	Use case
1× (default mobile)	32 × 32 px	Mobile portrait, full-world overview
2× (default desktop)	64 × 64 px	Desktop default, comfortable city detail
3× (zoomed)	96 × 96 px	Inspecting buildings, battles
4× (max)	128 × 128 px	Aesthetic appreciation, screenshots

4. Re-skinnable Sprite-Set Architecture

Conquer's sprites are organized in named **sets** so the visual style can be swapped without code changes. The default and only initial set is pixel (this document's pixelart style). Future sets could include hand-drawn, hd-pixel, or seasonal variants like pixel-winter.

Folder structure

```
assets/ ■■■ sprites/ ■■■ active.json # which sprite set(s) are enabled ■■■ pixel/ # the
default 32-color pixelart set ■ ■■■ manifest.json # version, palette ref, dimensions ■
■■■ terrain/ ■ ■ ■■■ grass_01.png ■ ■ ■■■ forest_01.png ■ ■ ■■■ ... ■ ■■■ monsters/ ■
■■■ buildings/ ■ ■■■ treasures/ ■ ■■■ charms/ ■ ■■■ ui/ ■ ■■■ cities/ ■■■ (future
sets here) ■■■ hand-drawn/ ■■■ ...
```

active.json — selecting the active sprite set

```
{ "default_set": "pixel", "available_sets": ["pixel"], "user_selectable": false
}
```

Phase 1 (MVP/launch): only pixel set exists, no user choice. Phase 2+: if a second set becomes available, players can choose in their settings.

manifest.json — per-set metadata

```
{ "name": "Pixel – 32-color SNES style", "version": "1.0.0", "author": "Sven
Manderscheid", "palette": "data/palette/conquer-32.gpl", "tile_size": 32,
"city_size": 64, "monster_size": 32, "boss_size": 64, "shrine_size": 128,
"scaling": "nearest-neighbor", "anti_aliasing": false, "completeness":
"incomplete", "missing_categories": ["treasures", "charms", "ui_advanced"] }
```

The renderer reads active.json on each request, resolves the active set, loads its manifest.json, and serves sprites from that folder. If a sprite is missing in the active set, fallback to the default set. This makes incomplete sets safe to ship.

5. Terrain Tiles

Terrain forms the base layer. Every map tile has a terrain background. Transitions between adjacent terrain types are handled by edge-tiles (autotiling, similar to RPG Maker).

Terrain	Variants	Total	Notes
Plains (default)	4 grass variations	4	Most common, ~50% of map
Forest	4 density variations	4	Lumber-rich biome
Mountains	4 rock variations	4	Stone-rich, blocks city placement
Water	1 base + 4 wave-anim frames	5	Impassable for marches
Desert	4 sand variations	4	Visual only, no gameplay effect
Snow	4 ice variations	4	Visual only
Edge Transitions	8 per terrain pair	~25	Plains↔Forest, Plains↔Water, etc.

Total: ~50 terrain sprites at 32×32 px

AI Prompt Template (Pixelart Terrain)

32x32 pixel art, 16-bit SNES style, [TERRAIN_TYPE], top-down game tile, limited 32-color palette, no anti-aliasing, crisp pixel edges, seamless tileable, soft natural shading, slight texture detail, fantasy strategy game aesthetic, clean and readable

Replace [TERRAIN_TYPE] with: lush green grass with small flowers, dense pine forest, rocky mountain stone, calm blue water with subtle ripples, golden desert sand with dunes, snow-covered tundra with ice patches, etc.

AI generation workflow

1. Generate at 256×256 with the prompt above (most AI tools work better at higher resolution).
2. Downscale to 32×32 with nearest-neighbor in Aseprite or Photoshop.
3. Reduce palette to the Conquer 32 colors (Aseprite: Sprite → Color Mode → Indexed → choose conquer-32.gpl).
4. Hand-clean 2-5 minutes per sprite — fix bad pixels, ensure tileability.
5. Save as PNG-32 with transparency.

6. Player Cities & Structures

Player City (4 Visual Tiers)

Each player city occupies a 2×2 area on the map. Castle level determines which of 4 visual tiers is shown. The city is what other players see when looking at the map — it should be readable and impressive at higher tiers.

Tier	Castle Level	Visual Description	Size
Tier 1	L1-5	Wooden palisade, single watchtower, simple cottages	64×64 px
Tier 2	L6-15	Stone walls, central keep, market stalls visible	64×64 px
Tier 3	L16-25	Larger castle with multiple towers, banners	64×64 px
Tier 4	L26-30	Endgame fortress, dramatic spires, elite aesthetic	64×64 px

AI Prompt Template (Pixelart Cities)

64x64 pixel art, 16-bit SNES style, [CITY_DESCRIPTION], top-down view fantasy strategy game, transparent background, limited 32-color palette, no anti-aliasing, crisp pixel edges, soft top-left lighting, [TIER_ACCENT_COLOR] accents, centered composition, no UI elements

Tier 1: simple wooden palisade settlement with single tower; Tier 4: grand fortified castle with multiple spires, fluttering banners, gold trim. Tier accent colors: T1=brown, T2=stone-grey, T3=blue-banner, T4=gold-and-red.

Conquest Structures

Structure	Count	Size	Visual
C-Shrine	20	128×128 px	Modest stone shrine, faint glow
B-Shrine	12	128×128 px	Larger temple, blue energy aura
A-Shrine	4	128×128 px	Imposing fortress-shrine, golden aura
Congress (S)	1	128×128 px	Massive central pyramid, brilliant divine glow
Alliance Flag (overlay)	—	32×32 px	Banner displayed on captured shrines

7. Monsters

Monsters need **two sprite versions**: a small map sprite (32×32) for the world view, and a larger battle sprite (64×64 standard, 96×96 for dragons, 128×128 for Magdar) for the battle report screen.

Monster	Levels	Map Size	Battle Size	Visual Tiers
Orc	Lv 0-10	32×32	64×64	4 tiers (L0-3, 4-6, 7-8, 9-10)
Skeleton	Lv 1-10	32×32	64×64	4 tiers
Golem	Lv 1-10	32×32	64×64	4 tiers
Treasure Goblin	Lv 1-5	32×32	64×64	2 tiers + glittering particle effect
Deathkar	Lv 1-10	32×32	64×64	4 tiers
Green Dragon	Lv 1-3	48×48	96×96	3 levels (color-shift)
Red Dragon	Lv 1-3	48×48	96×96	3 levels
Gold Dragon	Lv 1-3	48×48	96×96	3 levels
Magdar	Lv 1-3	64×64	128×128	3 levels (apex predator)

Total: ~30 map sprites + ~30 battle sprites = ~60 monster sprites

Pixelart Monster Style Suffix

Append this suffix to ALL monster prompts so they look cohesive:

, pixel art, 16-bit SNES style, top-down 2d game sprite, transparent background, 32-color palette, no anti-aliasing, crisp pixel edges, soft shading, single character centered, fantasy strategy game aesthetic, readable silhouette

Static Monster Prompts

Orc

32x32 pixel art, muscular green-skinned orc warrior, tribal armor with bone fragments, wielding crude iron axe, fierce snarl, low-tier: smaller and rough; high-tier (Lv 9-10): larger, scarred, ornate war-helm and gold-trimmed shield [PIXELART STYLE SUFFIX]

Skeleton

32x32 pixel art, undead skeletal warrior, tattered grey cloak, rusted armor pieces, glowing red eye-sockets, holding notched sword, low-tier: simple bones; high-tier: ornate dark armor and ethereal blue-purple aura [PIXELART STYLE SUFFIX]

Golem

32x32 pixel art, stone construct, jagged grey rock with glowing veins, hulking humanoid, moss patches, low-tier: small and rough; high-tier: massive with intricate runes carved into stone glowing amber [PIXELART STYLE SUFFIX]

Treasure Goblin

32x32 pixel art, small green goblin with oversized burlap sack overflowing with gold coins and gems, glittering sparkle particles, mischievous grin, large pointed ears, dagger in belt, golden crown askew, PRIMARY VISUAL CUE: bag of treasure must be obvious [PIXELART STYLE SUFFIX]

Deathkar

32x32 pixel art, skeletal undead horse rider, grim reaper aesthetic, tattered black robes, scythe weapon, mounted on bone-armored skeletal horse, hollow glowing eyes, ornate crown of bones, dark purple aura [PIXELART STYLE SUFFIX]

Green Dragon

48x48 pixel art, majestic emerald green dragon, large leathery wings spread, iridescent scales, yellow eyes, smoke from nostrils, perched on rocky outcrop, Lv 3: even larger, golden horn tips, deeper emerald glow [PIXELART STYLE SUFFIX]

Red Dragon

48x48 pixel art, fierce crimson red dragon, scales like burning coals, flames around mouth, muscular wings, sharp horns, glowing orange eyes, perched aggressively, Lv 3: imposing, lava cracks visible on scales [PIXELART STYLE SUFFIX]

Gold Dragon

48x48 pixel art, majestic golden dragon, brilliant metallic scales reflecting light, regal posture, bright blue eyes, elegant horns, divine aura, Lv 3: glowing brighter, ornate jeweled patterns [PIXELART STYLE SUFFIX]

Maqdar

64x64 pixel art, apex demonic dragon-king, massive dark obsidian scales with red-glowing fissures, four eyes glowing molten red, multiple horns, two pairs of wings, lava dripping from maw, ancient runic patterns on chest, throne-like rocky perch, Lv 3: visually overwhelming, surrounded by floating embers and ash [PIXELART STYLE SUFFIX]

8. Monster Animations (Pixelart)

Pixelart animations are **frame-by-frame** — typically 4-8 frames for a loop. This is one of pixelart's strengths: fewer frames look intentional rather than cheap, while HD animation expects 24-60 fps to feel smooth.

Animation specifications

Animation Type	Frames	Duration	FPS	Loop?	Used Where
Idle (map)	4 frames	1.0s	4	Yes	Map view
Idle (battle)	6 frames	1.5s	4	Yes	Battle screen
Attack	8 frames	0.8s	10	No	Battle on attack
Death	6 frames	0.6s + hold	10	No	Battle on defeat
Boss-Special (Magdal)	12 frames	1.5s	8	No	Phase transitions

Format: sprite sheets

Each animation is a horizontal sprite sheet (frames laid out left-to-right). Example: `orc_idle_sheet.png` at $128 \times 32 = 4$ frames of 32×32 . The renderer cycles through frames with CSS `background-position` animation or via Canvas `drawImage` with offset.

Tools for pixelart animation

Tool	Cost	Strength
Aseprite	\$20 once	BEST — built for pixelart animation, onion-skin, sprite sheets export
Piskel	Free (web)	Good for quick edits, browser-based, no install
LibreSprite	Free	Aseprite fork, slightly less polished but identical workflow
Pixilart	Free (web)	Easy beginner tool, online community
Krita	Free	Has pixel mode but less ergonomic than Aseprite

AI Animation Workflow

Pixelart animations rarely come out clean from AI tools — frame-to-frame consistency is hard for current models. **Recommended workflow:**

1. Generate the static sprite with AI (Stable Diffusion + pixelart prompt).
2. Clean it up in Aseprite.
3. **Animate by hand** — typically 4-8 frames where you nudge a few pixels per frame (breathing, eye blink, weapon sway, idle bob).
4. Export as horizontal sprite sheet PNG. Time per simple idle animation: ~15-30 minutes once you have the static frame.

Animation prompts (text descriptions for the artist)

Orc — Idle Animation

4 frames, 1s loop. Frames: (1) baseline pose, (2) chest rises 1 pixel – breathing in, (3) chest holds, (4) chest drops 1 pixel – breathing out. Optional: tribal beads

sway on frame 2-3. Axe stays still.

Skeleton — Idle Animation

4 frames, 1s loop. Frames: (1) baseline, (2) jaw drops 1 pixel – chatter, (3) eye-glow brighter (cycle palette index for socket pixels), (4) jaw closes. Cloak ripples 1-2 pixels at edges throughout.

Golem — Idle Animation

4 frames, 1s loop. Slow heavy breathing – chest rises 1 pixel over 2 frames, holds, drops over 2 frames. Glow veins pulse: brighter on frames 2-3, dimmer on 1-4. Add 1-pixel pebble fall from shoulder on frame 3 (animated dust).

Treasure Goblin — Idle Animation

6 frames, 1.5s loop. SPARKLE animation on bag is the focus – 3 sparkle pixel positions that cycle through frames (sparkle1 on frames 1-2, sparkle2 on 3-4, sparkle3 on 5-6). Goblin nervously glances side-to-side: eyes shift 1 pixel left on frame 2, right on frame 5. Crown wobbles 1 pixel on frame 4.

Treasure Goblin — Flee Animation

8 frames, 0.6s. Goblin transitions from idle to running pose, tightens grip on bag, spills 2-3 coin pixels behind. Used when player engages – telegraphs the despawn risk.

Deathkar — Idle Animation

6 frames, 1.5s loop. Horse paws ground: front leg lifts 2 pixels on frames 3-4. Rider raises and lowers scythe 2 pixels over the loop. Purple aura cycles brighter on frames 3-4, dimmer otherwise. Tattered cloak ripples 1 pixel at bottom edge.

Dragons (Green/Red/Gold) — Idle Animation

6 frames, 1.5s loop. Wings flex slightly: open 2 pixels wider on frames 2-4. Tail swishes: tip moves 1 pixel left on frame 3, right on frame 5. Smoke/flame from mouth: 2-pixel wisp on frames 1-3, dissipates on 4-6, then restart. Red Dragon: scales glow brighter on frames 3-4 (heat pulse). Gold Dragon: scales shimmer – cycle 2 pixels of bright→dim across body.

Magdar — Idle Animation (BOSS)

8 frames, 2s loop. Lava cracks pulse all over body: brighten across frames 1-4, dim across frames 5-8. All four eyes blink asynchronously: eye1 closes on frame 3, eye2 on frame 5, eye3 on frame 7, eye4 on frame 1. Wings flex 2 pixels wider on frames 4-5. Embers float upward: 3 ember-pixels rise 1 pixel per frame, fade out by frame 6, respawn on 7-8.

Magdar — Attack Animation (BOSS)

12 frames, 1.5s, no loop (fires once). Frames 1-3: charge up – eyes glow brighter, jaw opens. Frames 4-6: massive flame breath out (animate fire pixels expanding outward). Frames 7-9: flame dissipates. Frames 10-12: return to idle pose. Trigger screen-shake on frame 4 via JS, not in sprite.

9. Troops, Treasures, Charms & UI

Troop Sprites (Battle View)

Type	Tier 1	Tier 2	Tier 3	Tier 4	Tier 5
Infantry	Fighter	Warrior	Knight	Guardian	Crusader
Ranged	Hunter	Longbow Man	Ranger	Crossbow Man	Sniper
Cavalry	Stableman	Horseman	Heavy Cav	Iron Cav	Dragoon

15 troop sprites at 48×48 px each. Higher tier = better armor, more detail, higher-quality weapons. Tier 5 should look elite/legendary — use Gold palette for trim and accents.

AI Prompt Template (Pixelart Troops)

```
48x48 pixel art, fantasy [TROOP_TYPE] [TIER_DESCRIPTION], 16-bit SNES style, front-facing combat pose, weapon raised, transparent background, 32-color palette, no anti-aliasing, single character centered, soft top-left lighting, [ARMOR_QUALITY] armor, [DETAIL_LEVEL] detail
```

Example for Crusader (Inf T5): 'fantasy infantry knight in elite plate armor, ornate with gold trim and crusader insignia, wielding glowing two-handed sword, dramatic pose, noble stance, legendary tier' + style suffix.

Treasure Sprites

Grade	Count	Color Theme	Frame Style
Normal	14	Stone-grey palette	Plain stone border
Rare	15	Water-blue palette	Subtle blue glow border
Epic	26	Magic-purple palette	Magenta aura border
Legendary	16	Gold palette	Golden flames border
Mythic	6	Multi-color radiant	Multi-colored brilliance border

Total: 77 unique treasure sprites + 5 frame overlays at 48×48 px. Each treasure shows the item itself (e.g., 'Magic Plow' shows a glowing plow). Frame overlays sit on top of treasure images and indicate grade.

Charm Sprites

Charms are temporary stat-boost items. **24 designs total: 8 stat categories × 3 grades.** All 32×32 px.

Stat Category	Pixelart Visual Theme
Construction Speed	Pixel hammer + small gear icon, wood + grey palette
Research Speed	Open book with glowing rune, paper + magic palette
Troops HP	Heart shield, red + stone palette
Troops Attack	Crossed swords, stone + gold palette
Troops Defense	Tower shield, stone + gold palette

Carry Capacity	Saddlebag with coins, wood + gold palette
March Speed	Winged boot with motion lines, leather + UI cyan
Gathering Speed	Sickle / scythe, wood + grass palette

10. UI Icons & Elements

User interface icons in pixelart need to be readable at small sizes. Use bold silhouettes, avoid fine detail, and lean on the palette's distinctive colors for instant recognition.

Category	Items	Sizes	Total
Resources	Food, Lumber, Stone, Gold, GEMS	16×16 + 32×32 px	10
Speedups	Build / Research / Train / Recover / Generic	32×32 px	5
Action Buttons	Build, Research, Train, Attack, Spy, Gather, Rally, Heal, Trade	32×32 px	~20
March Type Icons	Gather / Attack / Support / Rally / Spy	24×24 px	5
Buff Icons	Production, Anti-Recon, Active Buffs	24×24 px	~15
Notification Icons	Battle Report, Spy Report, Mail, Alliance Help	16×16 px	~10
VIP Crowns	Levels 1-20 (or 4 tier-crowns)	32×32 px	4 (tiered) or 20
Avatar Frames	Default + 4 unlockable	48×48 px	5

Total UI Icons: ~85 items

UI Pixelart Conventions

Rule	Reason
Pixel-perfect at native size	Don't draw at 64×64 and downscale to 32×32 — draw at the target size
Bold 1-pixel outlines	Helps icons stand out on busy backgrounds
Distinctive color per category	Resources use their themed palette; Build = wood-brown, Research = magic-purple
Centered with 1-pixel padding	Leaves room for hover/active state borders
No gradients, no AA	Use 2-3 palette colors per icon — flat shading is the style

Buildings (Phase 2 City View)

MVP uses a list-based building view (cards with name, level, upgrade button). The full illustrated city view with isometric pixelart buildings is Phase 2 polish work.

Building	Tier 1 (L1-7)	Tier 2 (L8-15)	Tier 3 (L16-23)	Tier 4 (L24-30)
Castle	Wooden keep	Stone keep	Fortified castle	Endgame fortress
Wall	Wooden palisade	Stone wall	Reinforced battlements	Ornate fortifications
Farm	Small plot	Multi-field farm	Industrial farm	Magical bountiful farm
Lumber Camp	Single shed	Lumber yard	Logging operation	Mystical sawmill
Quarry	Open pit	Active quarry	Deep mine	Subterranean complex
Gold Mine	Mine entrance	Active gold mine	Industrial mine	Treasure vault
Storage	Wooden silo	Stone storehouse	Multi-warehouse	Crystal vault

Treasure House	Modest vault	Stone treasury	Ornate hall	Royal treasury
Barrack	Tents	Stone barracks	Training grounds	Royal academy
Hospital	Tent infirmary	Stone hospital	Healing temple	Magical sanctuary
Academy	Wooden lab	Library	University	Arcane spire
Trading Post	Market stall	Trading hall	Caravan center	Grand bazaar
Hall of Alliance	Meeting tent	Stone hall	Diplomatic center	Royal court

Total: 13 × 4 = 52 building sprites at 96×96 px (Phase 2)

11. Production Summary

Category	Sprites	Phase	Est. Hours (Solo Pixelartist)
Terrain (with transitions)	~50	MVP	20-35h
Player Cities (4 tiers)	4	MVP	6-10h
Conquest Structures	5	MVP	8-12h
Field Objects (3 tiers each)	15	MVP	10-18h
Monsters (Map + Battle)	60	MVP	40-70h
Monster Animations	~30 idle + 10 attack	MVP	30-60h
Troops (battle)	15	Phase 2	20-35h
Treasures	82	MVP	40-70h
Charms	24 + 3 frames	MVP	12-20h
UI Icons	~85	MVP	20-30h
Buildings (City View)	52	Phase 2	40-60h
Castle Skins (cosmetic)	5-10	Phase 4	8-20h

MVP subtotal: ~330 sprites — approximately 180-285 designer hours (less than HD because pixelart is faster per sprite)

Phase 2 add: +80 sprites — additional 60-95 hours

Phase 4 add: +5-10 sprites — additional 8-20 hours

Production tips

Tip	Reasoning
Do palette + 5 sample sprites first	1 terrain, 1 city, 1 monster, 1 building, 1 UI icon. Validate the style before scaling. Bad style direction v
Lock the palette early	Once 32-color palette is set, NEVER add more. The constraint is the feature.
Free starting points	Kenney.nl pixel packs (CC0), itch.io pixel asset bundles (\$5-20), OpenGameArt.org. Replace gradually
AI as draft tool	Stable Diffusion / Midjourney generate decent pixelart drafts. Always clean up in Aseprite — never ship
Sprite sheets after production	Pack into texture atlases (TexturePacker, Free-Tex-Packer). Reduces 330 HTTP requests to ~10. Big
Iterate per category	Build all terrain → see it on map → adjust → THEN do monsters. Don't generate all 330 sprites in para
Test mobile early	Pixelart at 32×32 on a 6" phone screen — test before committing 100 hours. If unreadable, switch to 4

12. File Organization

Folder structure (under assets/sprites/pixel/)

assets/sprites/pixel/ ■■■■ manifest.json # set metadata ■■■■ terrain/ ■ ■■■■ plains_01.png ... plains_04.png ■ ■■■■ forest_01.png ... forest_04.png ■ ■■■■ water_01.png + water_anim_sheet.png ■ ■■■■ transitions/ ■ ■ ■■■■ plains_to_forest_n.png (n=north edge) ■ ■ ■■■■ plains_to_water_n.png ■ ■ ■■■■ ... (8 directions × pairs) ■ ■■■■ ... ■■■■ cities/ ■ ■■■■ city_tier1.png ... city_tier4.png ■ ■■■■ skins/ ■ ■■■■ (Phase 4 skins) ■■■■ conquest/ ■ ■■■■ shrine_c.png ■ ■■■■ shrine_b.png ■ ■■■■ shrine_a.png ■ ■■■■ congress.png ■ ■■■■ alliance_flag.png ■■■■ field_objects/ ■ ■■■■ farm_t1.png farm_t2.png farm_t3.png ■ ■■■■ forest_t1.png ... ■ ■■■■ quarry_t1.png ... ■ ■■■■ gold_mine_t1.png ... ■ ■■■■ gem_node_lv1.png gem_node_lv2.png gem_node_lv3.png ■■■■ monsters/ ■ ■■■■ map/ ■ ■ ■■■■ orc_t1.png ... orc_t4.png ■ ■ ■■■■ ... ■ ■■■■ battle/ ■ ■ ■■■■ orc_battle_t1.png ... orc_battle_t4.png ■ ■ ■■■■ ... ■ ■■■■ animations/ ■ ■■■■ orc_idle_sheet.png ■ ■■■■ orc_attack_sheet.png ■ ■■■■ magdar_idle_sheet.png ■ ■■■■ ... ■■■■ troops/ ■ ■■■■ infantry_t1.png ... infantry_t5.png ■ ■■■■ ranged_t1.png ... ■ ■■■■ cavalry_t1.png ... ■■■■ treasures/ ■ ■■■■ normal/ (14 PNGs) ■ ■■■■ rare/ (15 PNGs) ■ ■■■■ epic/ (26 PNGs) ■ ■■■■ legendary/ (16 PNGs) ■ ■■■■ mythic/ (6 PNGs) ■ ■■■■ frames/ (5 grade-frame overlays) ■■■■ charms/ ■ ■■■■ construction_normal.png ... construction_legendary.png ■ ■■■■ ... (8 categories × 3 grades) ■ ■■■■ frames/ ■■■■ ui/ ■ ■■■■ resources/ ■ ■■■■ speedups/ ■ ■■■■ actions/ ■ ■■■■ buffs/ ■ ■■■■ notifications/ ■ ■■■■ vip/ ■ ■■■■ avatars/ ■■■■ buildings/ (Phase 2) ■■■■ atlases/ (packed sprite sheets – generated, not source)

Naming conventions

Element	Pattern	Example
Terrain	[type]_[variant].png	plains_01.png
Terrain transition	[type]_to_[type]_[direction].png	plains_to_forest_n.png
City	city_tier[1-4].png	city_tier3.png
Monster Map	[name]_t[1-4].png	orc_t2.png
Monster Battle	[name]_battle_t[1-4].png	magdar_battle_t3.png
Animation Sheet	[name]_[anim]_sheet.png	dragon_idle_sheet.png
Treasure	treasure_[grade]_[name].png	treasure_legendary_dragon_horn.png
Charm	charm_[stat]_[grade].png	charm_construction_legendary.png
UI	ui_[category]_[name].png	ui_resource_food_32.png

13. Sprint 0 Asset Plan

The bare minimum to validate the pixelart style works in-engine. Don't produce more until these are in the game and feel right.

Asset	Why it's first	Time
Conquer 32-color palette (.gpl)	Foundation for everything else	1-2h
6 base terrain tiles (4 grass + 1 forest + 1 water)	Map renders with placeholder colors otherwise	4-8h
1 player city (Tier 1)	Player can see their own city	2-4h
1 monster (Orc T1) — static + idle anim	Map has something to fight, validates animation pipeline	1-2h
10 UI icons (5 resources + 5 actions)	Game is interactable	3-5h
5 sample treasures (mixed grades)	Treasure system testable	3-5h

Sprint 0 total: ~16-29 hours = playable pixelart prototype

After Sprint 0: deploy these to assets/sprites/pixel/, render them in the game, take a screenshot, look at it on desktop AND mobile, decide if the style works. Iterate the palette and base sizes BEFORE producing 300 more sprites.